

Device for Maintaining Car Batteries



PETRE TZV. PETROV

When we do not drive our cars for extended periods of time, the abandoned rechargeable batteries become fully discharged. Sometimes, these get damaged, too.

Batteries have self-discharging rates that depend on their type and capacity, and may vary from a few milliamperes to tens of milliamperes. Also, the electrical installations in a car have leakage currents that may discharge the battery. If we compensate the discharging current, the battery will remain always charged. This can reduce the probability of the battery getting discharged or damaged. In any case, it is good to read

the datasheet of the manufacturer in order to know how to maintain the battery.

Circuit and working

Fig. 1 shows the simple device that can maintain two 12V rechargeable batteries. The mains power supply is applied to connector CON1 and is reduced by transformer X1 to 21V AC, 200mA. The bridge rectifier comprising diodes D1 through D4 rectifies the AC voltage.

The device has two identical channels. Each channel has an individual low-cost linear regulator 7805 (IC1 and IC2). Minimum output current for the battery is set by resistors R2 and R5 at around 10mA. Maximum output current can have two

values (around 50mA and 100mA).

When switches S2 and S3 are at position 1, the output current is around 50mA. When switches S2 and S3 are at position 2, the output current is around 100mA. Maximum output voltage is limited to around 15.5V with LED1, ZD1 and ZD2 for the first channel and LED2, ZD3 and ZD4 for the second channel.

Raising the current to around 100mA should not damage LED1 and LED2. Diodes D5 and D6 prevent the discharge of batteries BATT.1 and BATT.2 when the mains power supply is not present. Nevertheless, if the mains power supply is not present for a long time, disconnect the batteries from the device.

Mount regulators IC1 and IC2

on small individual heat-sinks with thermal resistance below 30°C/W. The device will start working immediately if the circuit is wired properly. Verify the maximum output voltage and maximum output current with a multimeter.

Construction and testing

An actual-size, single-side PCB for the power supply is shown in Fig. 2 and its component layout in Fig. 3. Enclose the PCB in a suitable small box in such a way that outputs can be easily taken from the front panel.

The ground pin of each 7805 IC is connected to metal-

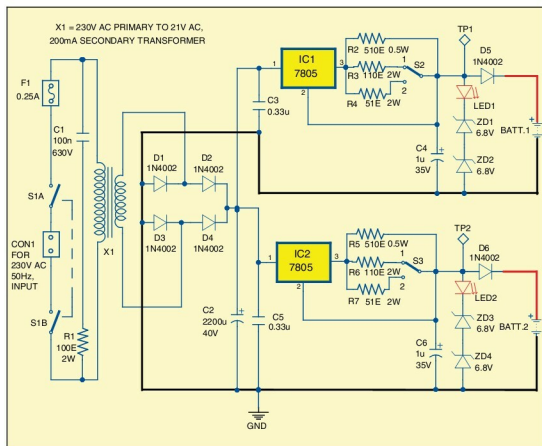


Fig. 1: Circuit of the simple device maintaining two 12V rechargeable batteries